

1) Name and EMPLID

- Ahmed Huossein
- 24417746

2) How much total time I spent

7 hours

3) How I managed development + testing

I developed using VSCode and stored changes in my code on GitHub. Any testing was done on the local terminal.

<https://github.com/ahmedhuossein/CSC-30400-Ahmed-Huossein-Turing-Machine-Project/tree/main>

4) Language + libraries I used

- Python
- Libraries used: sys

5) Key data structures

Each tape is stored as a Python **string**, and all tapes are kept inside a list (tapes). The position of each tape head is tracked using an **array** of integers (heads). The symbols currently under each head are collected into an **array** (current_symbols). States are represented simply as strings, including the start, accept, and reject states, and the full set of states is stored in an **array** (states). The transition function is stored in a **dictionary** (stored_transitions) keyed by rule number; each entry contains a dictionary specifying the current state, the list of symbols to read (one per tape), the next state, the list of symbols to write, and the list of head movements (“L”, “R”, or “S”)